

1 Groups and Fields

Definition 1. Group $(G, *)$ is a set G with a operation $*$ having the following properties

1. $G \neq \emptyset$: non-empty
2. $*$: $G \times G \rightarrow G$ is a function. We write $*((a, b))$ as $a * b$ for $a, b \in G$.
3. $\forall a, b, c \in G; a * (b * c) = (a * b) * c$: associative
4. $\exists e \in G, \forall a \in G; e * a = a * e = a$: identity exists
5. $\forall a \in G, \exists \bar{a} \in G; a * \bar{a} = \bar{a} * a = e$: inverse exists

Definition 2. Abelian Group $(G, *)$

- 1-5. $(G, *)$ is a group
6. $\forall a, b \in G; a * b = b * a$: commutative

Example 1. Check which of the following are groups

1. $(\mathbb{R}, +)$
2. $(\mathbb{R} - \{1\}, +)$
3. $(\mathbb{R}, \cdot)$
4. $(\mathbb{R} - \{0\}, \cdot)$
5. $GL_2(\mathbb{R})$ =General Linear group= all invertible 2×2 matrices with real entries, under matrix multiplication.
6. $\mathbb{B} = \{0, 1\}$ with boolean addition $+$
7. $\mathbb{B} = \{0, 1\}$ with boolean multiplication $\cdot$
8. $D_3 = \{R_0, R_1, R_2, R_3, L_1, L_2, L_3\}$ =Dihedral group=set of symmetries of an equilateral triangle, under composition.
9. Elliptic Curve with point at infinity: $\mathbb{E} = \{(x, y) | y^2 = x^3 + Ax + B, 4A^3 + 27B^2 \neq 0\} \cup \mathcal{O}$ under elliptic curve addition $+$: If P, Q, R on a straight line in $\mathbb{E}$ then $P + Q + R = \mathcal{O}$.

Theorem 1. If $(G, *)$ is a group and $a \in G$. Then

1. e is unique
2. $\bar{a}$ is unique
3. $\bar{\bar{a}} = a$

Definition 3. Field $(F, +, \cdot)$ is a set with two binary operations $+$ and $\cdot$ having the following properties

1. $(F, +)$ is an abelian group. We write $e = 0$ and $\bar{a} = -a$
2. $(F - \{0\}, \cdot)$ is an abelian group. We write $e = 1$ and $\bar{a} = a^{-1}$
3. $\forall a, b \in F; a \cdot b \in F$
4. $\forall a, b, c \in F; a \cdot (b + c) = (a \cdot b) + (a \cdot c)$: distributive

Example 2. Check which of the following are groups

1. $(\mathbb{R}, +, \cdot)$
2. $(\mathbb{R}, \cdot, +)$
3. $(\mathbb{Q}, +, \cdot)$
4. $(\mathbb{Z}, +, \cdot)$
5. $\mathbb{B} = \{0, 1\}$ with boolean addition and multiplication

6. $\mathbb{B} = \{0, 1\}$ with mod 2 addition and multiplication
7. $\{0, 1, 2, 3\}$ with mod 4 addition and multiplication
8. $(GL_2(\mathbb{R}), +, \cdot)$

Theorem 2. *Finite Fields*

1. If p is a prime, the set $\mathbb{F}_p = \{0, 1, 2, 3, \dots, p-1\}$ under mod p addition and multiplication is a field.
2. If F is a finite field with n number of elements, then $n = p^k$ for some prime p and integer k .

Theorem 3. *If $(F, +, \cdot)$ is a field and $a \in F$*

1. $a \cdot 0 = 0$
2. $1 \neq 0$
3. There must be at least two elements in a field.

Definition 4. *Vector space $(V, \#, *)$ over the field $(F, +, \cdot)$*

1. $(V, \#)$ is an abelian group
2. $(F, +, \cdot)$ is a field
3. $*$: $F \times V \rightarrow V$ is a function. We write $*((a, x))$ as $a * x$ for $a \in F$ and $x \in V$.
4. $\forall a, b \in F, \forall x \in V; (a + b) * x = (a * x) \# (b * x)$
5. $\forall a \in F, \forall x, y \in V; a * (x \# y) = (a * x) \# (a * y)$
6. $\forall a, b \in F, \forall x \in V; (a \cdot b) * x = a * (b * x)$
7. $\forall x \in V; 1 * x = x$

Note 1. *Vector Space operations*

$$+ : F \times F \rightarrow F$$

$$\cdot : F \times F \rightarrow F$$

$$\# : V \times V \rightarrow V$$

$$* : F \times V \rightarrow V$$

Example 3. *Check which of the following are vector spaces*

1. $(\mathbb{R}^3, +, \cdot)$ over $(\mathbb{R}, +, \cdot)$
2. $(\mathbb{R}^3, +, \cdot)$ over $(\mathbb{C}, +, \cdot)$
3. $(\mathbb{C}^3, +, \cdot)$ over $(\mathbb{R}, +, \cdot)$
4. $(\mathbb{C}^3, +, \cdot)$ over $(\mathbb{C}, +, \cdot)$
5. $(\mathbb{Q}_n[x], +, \cdot)$ over $(\mathbb{Q}, +, \cdot)$ where $\mathbb{Q}_n[x]$ is the set of polynomials of degree n or less in x with coefficients in $\mathbb{Q}$.
6. $(\mathbb{R}^{m \times n}, +, \cdot)$ over $(\mathbb{R}, +, \cdot)$ where $\mathbb{R}^{m \times n}$ is the set of $m \times n$ degree matrices with coefficients in $\mathbb{R}$.
7. $(\mathbb{R}^+, \#, *)$ over $(\mathbb{R}, +, \cdot)$ where $\#$ and $*$ operations are defined as $x \# y = xy$ and $a * x = x^a$ for $x, y \in \mathbb{R}^+$ and $a \in \mathbb{R}$

Theorem 4. *$(V, \#, *)$ over $(F, +, \cdot)$ is a vector space*

1. $\forall x \in V; 0 * x = e$
2. $\forall a \in F; a * e = e$
3. $\forall a \in F, \forall x \in V; a * x = e \Rightarrow a = 0$ or $x = e$
4. $\forall x \in V; (-1) * x = \bar{x}$

Note 2. When there is no confusion, we will use the following notation and names

1. $x \# y = x + y$: vector addition
2. $a * x = a \cdot x = ax$: scalar multiplication
3. $e = \underline{0}$
4. $\bar{x} = -x$
5. $(V, \#, *)$ over $(F, +, \cdot)$ will be written as $(V, +, \cdot)$ over $(F, +, \cdot)$ or V over F or just V when F is implied

Note 3. We will rewrite everything in the above notation for the vector space V over F

- 1.1 $V \neq \emptyset$
- 1.2 $+: V \times V \rightarrow V$ is a function, i.e. $+$ is a binary operation on V
- 1.3 $\forall x, y, z \in V; x + (y + z) = (x + y) + z$
- 1.4 $\exists \underline{0} \in V, \forall x \in V; x + \underline{0} = \underline{0} + x = x$
- 1.5 $\forall x \in V; \exists -x \in V; x + (-x) = (-x) + x = \underline{0}$
- 1.6 $\forall x, y \in V; x + y = y + x$
2. $(F, +, \cdot)$ is a field
3. $\cdot: F \times V \rightarrow V$ is a function
4. $\forall a \in F, \forall x, y \in V; a \cdot (x + y) = (a \cdot x) + (a \cdot y)$
5. $\forall a, b \in F, \forall x \in V; (a + b) \cdot x = (a \cdot x) + (b \cdot x)$
6. $\forall a, b \in F, \forall x \in V; (a \cdot b) \cdot x = a \cdot (b \cdot x)$
7. $\forall x \in V; 1 \cdot x = x$

Definition 5. Subspace

S is a subspace of the vector space V over F iff S is a non empty subset of V and S is a vector space over F .

Example 4. Which of the following are subspaces of the given vector space

1. $(\mathbb{R}^3, +, \cdot)$ of $(\mathbb{C}^3, +, \cdot)$ over $(\mathbb{R}, +, \cdot)$
2. $(\mathbb{R}_3[x], +, \cdot)$ of $(\mathbb{R}[x], +, \cdot)$ over $(\mathbb{R}, +, \cdot)$
3. $(GL_2(\mathbb{R}), +, \cdot)$ of $(\mathbb{R}^{2 \times 2}, +, \cdot)$ over $(\mathbb{R}, +, \cdot)$
4. Set of solutions to the ODE $y'' - y - 6y = 0$ of Continuous functions $\mathcal{C}$ over $\mathbb{R}$.

Theorem 5. Let S be a non empty subset of the vector space V over F . If S is closed under vector addition and scalar multiplication, then S is a subspace of V over F .

Definition 6. Linear Combination

Let $B = \{x_1, x_2, \dots, x_n\}$ be a non empty finite subset of a vector space V over F . Then an element of the form $\sum_{k=1}^n a_k x_k = a_1 x_1 + a_2 x_2 + \dots + a_n x_n$ with $a_k \in F$ is called a Linear Combination of B .

Definition 7. Span

Let $B = \{x_1, x_2, \dots, x_n\}$ be a non empty finite subset of the vector space V over F . Then the collection of all possible linear combinations is called the Span of B .

i.e. $\text{Span} B = \{\sum_{k=1}^n a_k x_k \mid x_k \in B, a_k \in F\}$

If $\text{Span} B = W$, we say that B spans W .

Example 5. Which of the following sets span $\mathbb{R}^3$ over $\mathbb{R}$?

1. $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
2. $B = \{(1, 0, 0), (0, 1, 0)\}$.
3. $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$.

Theorem 6.

1. $B \subset \text{Span} B \subset V$
2. $\underline{0} \in \text{Span} B$
3. $\text{Span} B$ is a Subspace of V .
4. $\text{Span} B$ is the smallest subspace containing B

Definition 8. Linear Independence

Let $B = \{x_1, x_2, \dots, x_n\}$ be a non empty finite subset of a vector space V over F . B is Linear Independent on V over F iff

$$\forall (a_1, a_2, \dots, a_n) \in F^n; \sum_{k=1}^n a_k x_k = \underline{0} \Rightarrow \forall k, a_k = 0$$

B is Linearly Dependent iff it is NOT Linearly Independent

i.e. $\exists (a_1, a_2, \dots, a_n) \in F^n; \sum_{k=1}^n a_k x_k = \underline{0}$ and $\exists k, a_k \neq 0$

Example 6. Which of the following sets are Linearly Independent in $\mathbb{R}^3$ over $\mathbb{R}$?

1. $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
2. $B = \{(1, 0, 0), (0, 1, 0)\}$.
3. $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$.

Example 7.

1. If B contains $\underline{0}$, can B be Linearly Independent?
2. Is $B = \{x\}, x \neq \underline{0}$ Linearly Independent?
3. Is the null set $\phi = \{\}$ is Linearly Independent?
4. Let B be Linearly Independent and $x \notin B$. Is $B \cup \{x\}$ Linearly Independent?
5. B is Linear Independent and $x \notin \text{Span} B$. Is $B \cup \{x\}$ Linearly Independent?
6. Is any subset of a LI set is LI?
7. Is any superset of a LD set is LD?

Theorem 7. If Let $B = \{x_1, x_2, \dots, x_n\}$ be linearly Dependent, then at least one element can be written as a linear combination of others.

Definition 9. Let B be a non empty finite subset of a vector space V over F . B is called a Basis iff B spans V and B is linearly independent.

Example 8. Which of the following sets are a basis for $\mathbb{R}^3$ over $\mathbb{R}$?

1. $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
2. $B = \{(1, 0, 0), (0, 1, 0)\}$.
3. $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$.
4. $B = \{(1, 0, 0), (0, 1, 0), (1, 1, 1)\}$.
5. $B = \{(1, 0, 0), (0, 0, 1), (1, 1, 1)\}$.
6. $B = \{(0, 1, 0), (0, 0, 1), (1, 1, 1)\}$.

Example 9. Find two different basis for the following

1. $(\mathbb{Q}_n[x], +, \cdot)$ over $(\mathbb{Q}, +, \cdot)$ where $\mathbb{Q}_n[x]$ is the set of polynomials of degree n or

less in x with coefficients in $\mathbb{Q}$.

2. $(\mathbb{R}^{m \times n}, +, \cdot)$ over $(\mathbb{R}, +, \cdot)$ where $\mathbb{R}^{m \times n}$ is the set of $m \times n$ degree matrices with coefficients in $\mathbb{R}$.

3. Set of Solutions to the ODE $y'' - y - 6y = 0$ over $\mathbb{R}$.

Theorem 8. Let a basis B has n elements in a vector space V over the field F . Then

1. Any subset of V with more than n elements is LD.

2. Any subset of V with less than n elements does not span V .

Note 4. Solutions to $A_{n \times m} X_{m \times 1} = b_{n \times 1}$, i.e. n equations in m variables. If

1. $\text{rank}(A|b) > \text{rank} A$ then no solutions

2. $\text{rank}(A|b) = \text{rank} A = r$

2.1 $r = m$ then unique solution

2.2 $r < m$ then infinitely many solutions

2.3 $r > m$ is not possible

3 $\text{rank}(A|b) < \text{rank} A$ is not possible

Theorem 9. Any two bases in a vector space has the same number of elements.

Theorem 10. Representation of any element in a vector space by a given basis is unique. i.e. the associated field elements are unique.

Theorem 11. If B spans a vector space V but no proper subset of B spans V . Then B is a basis for V .

Theorem 12.

1. Let B be a LI subset of V . Then B can be extended to a basis.

2. Let B be a spanning subset of V . Then B can be reduced to a basis.

Theorem 13. B is a basis iff B is maximally linearly independent (no super set that contains B is linearly independent).

Example 10. Discuss bases for the following vector spaces

1. $(\mathbb{R}[x], +, \cdot)$ over $(\mathbb{R}, +, \cdot)$

2. $(\mathbb{R}, +, \cdot)$ over $(\mathbb{Q}, +, \cdot)$

3. $(\mathcal{C}[a, b], +, \cdot)$ over $(\mathbb{R}, +, \cdot)$

4. $(\mathcal{C}^\infty, +, \cdot)$ over $(\mathbb{R}, +, \cdot)$: Taylor Series

5. $(\mathbb{R}^\infty, +, \cdot)$ over $(\mathbb{R}, +, \cdot)$ with $(a_k) \in \mathbb{R}^\infty$ implies $\lim_{k \rightarrow \infty} a_k \in \mathbb{R}$: Converging real sequences.

6. Fourier Series of functions in $L_2[-\pi, \pi]$ i.e. in square integrable functions on $[\pi, \pi]$

Definition 10. A Hamel Basis B is a Basis Such that it need not be finite and only finite linear combinations are taken.

Definition 11.

1. A collection of sets C is called a Chain iff for all $A, B \in C$ either $A \subset B$ or $B \subset A$.

2. An element M of a collection of sets D is called Maximal iff M is not properly contained in any element of D .

Theorem 14. *Zorn's Lemma*

If for every chain C in D , there is U in D such that U contains every member of C (i.e. U is an upper bound of C), then D has a maximal element.

Theorem 15. *Every vector space has a Hamal Basis.*

Proof. Let D be the collection of all L.I. subsets of V . Let C be a chain in V . Consider $U = \cup_{C_i \in C} C_i$. Now U contains every element of C . We want to show that U is also in D .

Let $a_1x_1 + a_2x_2 + \cdots + a_nx_n = \underline{0}$, $x_k \in U$, $a_k \in F$ for $k = 1, 2, \dots, n$. Now $x_k \in C_k$ where C_k in C for $k = 1, 2, \dots, n$. But $C_1 \cup C_2 \cup \cdots \cup C_n = C_m$ for some $C_m \in C$. Now we have $C_m \in D$, i.e. C_m is L.I. and that $x_1, x_2, \dots, x_n \in C_m$. Therefore $a_1 = a_2 = \cdots = a_n = 0$. This implies that U is in D . Now we have shown that for every chain C in D , there is U in D such that U contains every element of C . By Zorns's lemma, there is a maximal element M in D . i.e. there is a set M such that M is L.I. and M is not contained in any L.I. set (M is maximally L.I.). Therefore M is a basis of V . $\square$

Definition 12. *Dimension*

No of elements of a basis of a vector space V is called its dimension or $\dim V$.

We define $\dim\{\underline{0}\} = 0$.

Note 5. Let $f(x) \in f \in \mathcal{L}_2[-\pi, \pi]$, i.e. $\|f\|_2^2 = \int_{-\pi}^{\pi} |f(x)|^2 dx \in \mathbb{R}$

1. Complex Fourier Series: $S(f) = \sum_{k=-\infty}^{\infty} c_k e^{ikx}$,

$$c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx$$

2. Real Fourier Series: $S(f) = c_0 + \sum_{k=0}^{\infty} a_k \cos kx + \sum_{k=0}^{\infty} b_k \sin kx$,

$$c_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx, a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx dx, b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx dx$$

Definition 13. A Schauder Basis B is a Basis Such that it is countably infinite and infinite sums are taken.

Theorem 16. If $f \in \mathcal{C}[-\pi, \pi]$ with $\sum_{k=-\infty}^{\infty} |c_k|$ is converging, then the Fourier Series exists and converging uniformly. Note that $f \in \mathcal{C}[-\pi, \pi] \Rightarrow f \in \mathcal{L}_2[-\pi, \pi]$.

Definition 14. *Inner Product*

Let V be a vector space over an Ordered Field $(F, +, \cdot, \leq)$. Then an Inner Product $\langle, \rangle$ on V is such that for all $x, y \in V$ and $a \in F$

1. $\langle, \rangle: V \times V \rightarrow F$ is a function

2. $\langle x, y \rangle = \langle y, x \rangle$

3. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$

4. $\langle ax, y \rangle = a \langle x, y \rangle$

5. $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ iff $x = \underline{0}$

Note that $F = \mathbb{C}$ is not an ordered field. But a proper subset $\mathbb{R}$ is an ordered field and we define/make sure

2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$

$$4. \langle ax, y \rangle = \bar{a} \langle x, y \rangle$$

$$5. \langle x, x \rangle, 0 \in \mathbb{R}$$

We define the norm of x as $\|x\| = \sqrt{\langle x, x \rangle}$

Theorem 17.

$$1. \langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$$

$$2. \langle x, ay \rangle = a \langle x, y \rangle$$

$$3. \langle x, \underline{0} \rangle = \langle \underline{0}, x \rangle = 0$$

Example 11. Show that the following are inner products

$$1. \langle x, y \rangle = x \cdot y = x^T y \text{ for } V = \mathbb{R}^n \text{ over } F = \mathbb{R}$$

$$2. \langle x, y \rangle = x^H \cdot y \text{ for } V = \mathbb{C}^n \text{ over } F = \mathbb{C}, \text{ where } x \text{ Hermitian is } x^H = \bar{x}^T = \overline{x^T}$$

$$3. \langle f, g \rangle = \int_0^\infty e^{-x} f(x) g(x) dx \text{ for } V = \mathbb{R}[x] \text{ over } F = \mathbb{R}$$

$$4. \langle A, B \rangle = \text{Tr}(A^H B) \text{ for } V = \mathbb{C}^{n \times n} \text{ over } F = \mathbb{C}, \text{ where Trace, Tr is the sum of diagonal elements.}$$

Theorem 18. Cauchy-Schwarz Inequality $|\langle u, v \rangle| \leq \|u\| \|v\|$

Theorem 19. Triangular Inequality $\|u + v\| \leq \|u\| + \|v\|$

Example 12. Investigate the Cauchy-Schwarz Inequality in $\mathbb{R}^n, \mathbb{C}^n, \mathcal{L}_2[a, b]$

Definition 15. $x, y \in V$

$$1. x, y \in V \text{ is orthogonal iff } \langle x, y \rangle = 0$$

$$2. x, y \in V \text{ is orthonormal iff } x, y \text{ is orthogonal and } \|x\| = \|y\| = 1$$

If the above is satisfied for all $x, y \in B$, we say that the set B is orthogonal or orthonormal respectively.

Theorem 20. $\underline{0} \notin B$ and B is orthogonal then B is linearly independent.

Theorem 21. Gram-Schmidt Orthogonalization

If $B = \{x_1, x_2, \dots\}$ be linearly independent then the set $C = \{y_1, y_2, \dots\}$ is orthonormal with

$$y_k = x_k - \sum_{j=1}^{k-1} \frac{\langle x_k, y_j \rangle}{\|y_j\|^2} y_j = x_k - \sum_{j=1}^{k-1} \langle x_k, \hat{y}_j \rangle \hat{y}_j$$

Example 13. Convert each of the following sets to an orthonormal set in two different ways using the inner products given above.

$$1. B = \{(1, 0, 0), (0, 1, 0), (1, 1, 1)\}.$$

$$2. B = \{1, x, x^2, x^3\}$$

Note 6. Let $V = \mathbb{R}[x], F = \mathbb{R}$ and $B = \{1, x, x^2, x^3, \dots\}$ be the standard basis.

$$1. \text{ Orthogonal basis w.r.t. the inner product } \langle f, g \rangle = \int_{-1}^1 f(x) g(x) dx \text{ is}$$

$$B = \{1, x, \frac{1}{2}(3x^2 - 1), \dots\} = \{P_n(x)\}_{n=0}^\infty \text{ where } P_n(x) \text{ are the Legendre polynomials.}$$

They satisfy the ODE: $(1 - x^2)y'' - 2xy' + n(n+1)y = 0$.

$$2. \text{ Orthogonal basis w.r.t. the inner product } \langle f, g \rangle = \int_0^\infty e^{-x} f(x) g(x) dx \text{ is}$$

$$B = \{1, -x + 1, \frac{1}{2}(x^2 - 4x + 2), \dots\} = \{L_n(x)\}_{n=0}^\infty \text{ where } L_n(x) \text{ are the Laguerre polynomials.}$$

They satisfy the ODE: $xy'' + (1 - x)y' + ny = 0$.

3. Orthogonal basis w.r.t. the inner product $\langle f, g \rangle = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} f(x)g(x)dx$ is $B = \{1, x, 2x^2 - 1, \dots\} = \{T_n(x)\}_{n=0}^\infty$ where $T_n(x)$ are the Chebyshev polynomials. They satisfy the ODE: $(1-x^2)y'' - xy' + n^2y = 0$. It is also true that $T_n(\cos \theta) = \cos n\theta$

Definition 16. Let W be a non empty subset of an inner product space V over F . The Normal Subspace of W , or the W perpendicular space is defined as $W^\perp = \{u \in V \mid \langle u, w \rangle = 0, \forall w \in W\}$

Theorem 22.

1. $W^\perp$ is a subspace of V .
2. $(W^\perp)^\perp \supseteq W$
3. $W \cap W^\perp = \{0\}$, if W also a subspace
4. If W also a subspace on V then any $u \in V$ can be written as $u = Pu + Qu$ where $Qu \in W^\perp$ and $Pu = \sum_i \langle u, w_i \rangle w_i \in W$. Here $\{w_i\}$ is an orthonormal basis for W . Using (3), we can show that this expression is unique.
5. In a setting described by (4), we write $V = W \oplus W^\perp$, and say that V is a direct sum of W and $W^\perp$.
6. With the setting in (4), Pu is the best approximation to u in W , i.e. $\|u - Pu\| \leq \|u - w\|$ for any $w \in W$.

Example 14.

1. Find the best approximation to $(1, 1, 1)$ in $W = \text{span}\{(1, 0, 0), (0, 1, 0)\}$.
2. Find the best approximation to $e^{\frac{x}{4}}$ in $W = \text{span}\{1, x, x^2, x^3, x^4\}$, use the inner product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx$

Definition 17. A norm $\|\cdot\|$ on a vector space V over $F = \mathbb{C}$ is such that for all $u, v \in V$ and $a \in F$

1. $\|\cdot\| : V \rightarrow F$ is a function
2. $\|v\| \geq 0$ and $\|v\| = 0$ iff $v = 0$
3. $\|av\| = |a|\|v\|$
4. $\|u + v\| \leq \|u\| + \|v\|$

If V also has a multiplication (like in the case of matrices), we also require that

5. $\|uv\| \leq \|v\|\|v\|$

Note 7.

1. If we have a norm, we can define Cauchy Sequences. That is, a sequence u_n is Cauchy iff $\forall \epsilon > 0 \exists N \in \mathbb{Z}^+ \forall m, n \in \mathbb{Z}^+; m, n > N \Rightarrow \|u_n - u_m\| < \epsilon$.
2. If All Cauchy sequences defined on V , converge to a point in V , we say that V is Complete.
3. A Banach Space is a Complete normed space.
4. As defined earlier, we can have a norm, that does not come from an inner product. But as you already know, if have an inner product, we have a Norm defined by $\|x\| = \sqrt{\langle x, x \rangle}$.
5. A Hilbert Space is a Complete inner product space.
6. If V is a Normed Space we have a metric (distance measuring function) defined by $d(x, y) = \|x - y\|$.

7. But we can have a matrix that does not come from norm. A general metric is defined as follows.

Definition 18. A metric d on a set V is such that for all $x, y, z \in V$ we have

1. $d : V \times V \rightarrow \mathbb{R}$ is a function
2. $d(x, y) = d(y, x)$
3. $d(x, z) \leq d(x, y) + d(y, z)$
4. $d(x, y) \geq 0$ and $d(x, y) = 0$ iff $x = y$.

Definition 19. Linear Transformation

Let V, W be two vector spaces over the common field F . Then a linear transformation T from V to W is such that for all $u, v \in V$ and $a \in F$

1. $T : V \rightarrow W$ is a function.
2. $T(u + v) = T(u) + T(v)$
3. $T(au) = aT(u)$

Example 15. Show that the following are linear transformations

1. $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4, F = \mathbb{R}$

$$T((x, y, z)) = (x - y + 2z, 2x + y, -x - 2y + 2z, 2x + y)$$

2. $T : \mathcal{C}[a, b] \rightarrow \mathbb{R}, F = \mathbb{R}, T(f) = \int_a^b f(x)dx$

3. $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}_2[x], F = \mathbb{R}, T(p(x)) = xp''(x) + p'(x)$

Theorem 23.

1. $T(\underline{0}) = \underline{0}$
2. $T(-u) = -T(u)$

Definition 20. Let $T : V \rightarrow W$ is a L.T. over F .

1. $\ker T = \{u \in V | T(u) = \underline{0}\}$: kernel
2. $\text{ran} T = \{T(u) | u \in V\}$: range

Theorem 24. Let $T : V \rightarrow W$ is a L.T. over F .

1. $\ker T$ is a subspace of V .
2. $\text{ran} T$ is a subspace of W .

we define

$$\text{null} T = \dim(\ker T)$$

$$\text{rank} T = \dim(\text{ran} T)$$

Example 16.

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^4, F = \mathbb{R}$$

$$T((x, y, z)) = (x - y + 2z, 2x + y, -x - 2y + 2z, 2x + y)$$

Find $\text{null} T$ and $\text{rank} T$

Theorem 25. $\text{null} T + \text{rank} T = \dim V$

Proof. Let $\dim(\ker T) = m$ and $B = \{u_1, u_2, \dots, u_m\}$ be a basis of $\ker T$. Let $\dim(V) = m + n$ and we can extend B to a basis $B \cup C$ of V by adding $C = \{v_1, v_2, \dots, v_n\}$ L.I. elements to B and all of them must be outside $\ker T$, since $\ker T$

can't contain more than m L.I. elements.

Now we claim that $D = \{T(v_1), T(v_2), \dots, T(v_n)\}$ is a basis for $\text{ran}T$:

If $a_1T(v_1) + a_2T(v_2) + \dots + a_nT(v_n) = \underline{0}$ for $a_k \in F$, we have

$T(a_1v_1 + a_2v_2 + \dots + a_nv_n) = \underline{0}$, which implies that $v = a_1v_1 + a_2v_2 + \dots + a_nv_n \in \ker T$.

Now since B is a basis for $\ker T$ we have $v = a_1v_1 + a_2v_2 + \dots + a_nv_n = b_1u_1 + b_2u_2 + \dots + b_mu_m$ for some $b_j \in F$. This implies

$a_1v_1 + a_2v_2 + \dots + a_nv_n - b_1u_1 - b_2u_2 - \dots - b_mu_m = \underline{0}$ and that $a_1 = a_2 = \dots = a_n = -b_1 = -b_2 = \dots = -b_m = 0$ since $B \cup C$ is a basis for V , hence L.I. This follows that $a_1 = a_2 = \dots = a_n = 0$ and that C is L.I.

Now for $T(u) \in \text{ran}T$ we have, $T(u) = T(a_1v_1 + a_2v_2 + \dots + a_nv_n + b_1u_1 + b_2u_2 + \dots + b_mu_m) = a_1T(v_1) + a_2T(v_2) + \dots + a_nT(v_n) + b_1T(u_1) + b_2T(u_2) + \dots + b_mT(u_m) = a_1T(v_1) + a_2T(v_2) + \dots + a_nT(v_n)$ since $T(u_k) = 0$, because $u_k \in \ker T$. This shows that $\text{span}D = \text{ran}T$. Since we already proved that D is L.I., this follows that D is indeed a basis for $\text{ran}T$ and that $\dim(\text{ran}T) = n$. Finally, $\dim(V) = m + n = \dim(\ker T) + \dim(\text{ran}T)$. $\square$

Example 17. (Related to Quiz9) Verify the rank-nullity theorem for $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}_2[x]$, $F = \mathbb{R}$, $T(p(x)) = xp''(x) + p'(x)$.

Proof. If $T(p(x)) = xp''(x) + p'(x) = 0$ for all x , we have, $xp''(x) = -p'(x)$ or $\frac{p''(x)}{p'(x)} = -\frac{1}{x}$. By integrating we have, $\log p'(x) = -\log x + \log a$ or $p'(x) = \frac{a}{x}$. Integrating again we have, $p(x) = a \log x + b$ where $a, b \in \mathbb{R}$. Since $p(x)$ must be in $\mathbb{R}_2[x]$ the only possibility is $a = 0$, and therefore $p(x) = b = b(1) \in \text{span}\{1\}$. Therefore $\ker T = \mathbb{R} = \text{span}\{1\}$. Since $\{1\}$ is L.I., it is a basis for $\ker T$ and $\dim(\ker T) = 1$.

Also for any $p(x) = a_3x^3 + a_2x^2 + a_1x + a_0$ we have $T(p(x)) = xp''(x) + p'(x) = x(6a_3x + 2a_2) + (3a_3x^2 + 2a_2x + a_1) = 9a_3x^2 + 4a_2x + a_1(1) \in \text{span}\{x^2, x, 1\}$. This shows that $\text{ran}T \subset \text{span}\{x^2, x, 1\}$. Now for any $ax^2 + bx + c \in \text{span}\{x^2, x, 1\}$ be an element of $\text{ran}T$, we need $ax^2 + bx + c = 9a_3x^2 + 4a_2x + a_1$. We can set this by letting $9a_3 = a, 4a_2 = b, a_1 = c$ or by $a_3 = a/9, a_2 = b/4, a_1 = c$. This shows that $\text{span}\{x^2, x, 1\} \subset \text{ran}T$. Therefore we have $\text{ran}T = \text{span}\{x^2, x, 1\}$. Since $\{x^2, x, 1\}$ is L.I., it is also a basis for $\text{ran}T$ and $\dim(\text{ran}T) = 3$.

We also see that $\{x^3, x^2, x, 1\}$ is a basis for $V = \mathbb{R}_3[x]$ and that $\dim(V) = 4 = 1 + 3 = \dim(\ker T) + \dim(\text{ran}T)$. $\square$

Definition 21. Matrix of a L.T.

Let $T : V \rightarrow W$ be a L.T. over F with $(u_i)_{i=1}^n$ and $(w_j)_{j=1}^m$ be ordered bases of V and W respectively. Then the matrix $A = T_{(u_i), (w_j)} \in F^{m \times n}$ such that $T((u_i)) = (w_j)A$ is called the Matrix of T w.r.t. $(u_i), (w_j)$.

Theorem 26. Let $(u_i), (u'_i)$ and $(w_j), (w'_j)$ be two ordered bases of V and W respectively. With $(u'_i) = (u_i)P$ and $(w'_j) = (w_j)Q$, and $A = T_{(u_i), (w_j)}, A' = T_{(u'_i), (w'_j)}$, we have $AP = QA'$ or $A' = Q^{-1}AP$.

Example 18. Find the matrix of A for each of the following L.T.s

1. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, F = \mathbb{R}, T((x, y)) = (x + y, x - 2y, 2x + 3y)$ w.r.t standard bases.
2. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, F = \mathbb{R}, T((x, y)) = (x + y, x - 2y, 2x + 3y)$ w.r.t $(u'_i) = ((1, 2), (1, 1))$ and $(w'_j) = ((1, 1, 0), (1, 0, 2), (0, 3, 3))$.
3. $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}_3[x], F = \mathbb{R}, T(p(x)) = xp''(x) + p'(x)$ w.r.t. standard bases
4. $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}_2[x], F = \mathbb{R}, T(p(x)) = xp''(x) + p'(x)$ w.r.t. $(u'_i) = (1, 1 + x, 1 + x + x^2, 1 + x + x^2 + x^3)$ and $(w'_j) = (1, 1 + x, 1 + x + x^2)$

Definition 22. $A \in F^{m \times n}$

1. Reduced row echelon form of $A = \text{rref}(A)$.
2. Row space of $A = \text{space spanned by rows of } A$.
3. Row rank of $A = \dim(\text{row space of } A)$.
4. Column space of $A = \text{space spanned by columns of } A$.
5. Column rank of $A = \dim(\text{column space of } A)$.

Theorem 27.

row rank of $A = \text{rank}(\text{row space of } A) = \text{rank}(\text{rref}(A)) = \text{rank}(\text{column space of } A) = \text{column rank of } A = \text{rank } A$.

Definition 23. Let $T : V \rightarrow V$ be a vector space over F . $\lambda \in F$ is a eigenvalue and $\underline{0} \neq v \in V$ is a corresponding eigenvector iff $T(v) = \lambda v$.

Eigen space V_λ of the eigen value λ are the set of all the eigen vectors corresponding to λ together with $\{0\}$.

i.e. $V_\lambda = \{v \in V | T(v) = \lambda v\} = \{v \in V | (T - \lambda I)(v) = \underline{0}\} = \ker(T - \lambda I)$, where I is the identity linear transformation $I : V \rightarrow V$ with $I(x) = x$ for all $x \in V$.

Note that if λ is existing, V_λ is a subspace of V . We can call its basis elements(which are non-zero) the eigenvectors.

Example 19. Find the eigenvalues and vector of the following L.T.s

1. $T : \mathcal{C}[a, b] \rightarrow \mathcal{C}[a, b], F = \mathbb{R}, T(f(x)) = \int_a^x f(t)dt$. Here eigenvalues/vectors DNE.
2. $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3, F = \mathbb{R}, T((x, y, z)) = (2x + y, -x - 2y + 2z, 2x + y)$
3. $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}_3[x], F = \mathbb{R}, T(p(x)) = xp'(x) + p(x)$
4. $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}_2[x], F = \mathbb{R}, T(p(x)) = xp''(x) + p'(x)$
5. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, F = \mathbb{R}, T(x) = Ax$,

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

5. $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3, F = \mathbb{R}, T(x) = Ax$,

$$A = \begin{pmatrix} 5 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 3 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

Definition 24.

1. Eigenvalues λ and Eigen Vectors v of the matrix $A \in F^{n \times n}$ is defined to be the same for the Linear Transformation $T : \mathbb{F}^n \rightarrow \mathbb{F}^n$ over F defined by $T(v) = Av$ where $v \in F^n$. i.e. for $A \in F^{n \times n}$, they satisfy $Av = \lambda v$ such that $\lambda \in F$ and $\underline{0} \neq v \in F^n$
2. $\det(A - \lambda I) = 0$.

3. $f(x) = \det(xI - A) = (-1)^n \det(A - xI)$, $x \in F$ is called the characteristic polynomial of A .
4. A, B are similar iff $AP = PB$ some invertible matrix P

Theorem 28.

1. Similar matrices have the same eigen values.
2. Eigen values of $T : V \rightarrow V$ over F and the eigen values of the matrix of T are the same for any choice of basis.
3. If the eigen values are different, then the corresponding eigen vectors are linearly independent.

Theorem 29.

1. $\prod \lambda = \det A$: product of eigenvalues is the determinant of A .
2. $\sum \lambda = \text{trace} A$: sum of eigenvalues is the trace of A (sum of the diagonal elements).
3. $f(A) = O$: matrix A satisfies its own characteristic polynomial (Cayley-Hamilton theorem).

Definition 25. Minimal polynomial $g(x)$ of $A \in \mathbb{C}^{n \times n}$ satisfies

1. $g(x)$ is monic
2. $g(A) = O$
3. no non-zero polynomial $h(x)$ with lesser degree satisfies $h(A) = O$.

Theorem 30.

1. $f(x)$ and $g(x)$ have the same roots.
2. $g(x)$ divides $f(x)$.

Definition 26.

1. Algebraic multiplicity of $\lambda = a_\lambda =$ power of $(x - \lambda)$ in $f(x)$.
2. Geometric multiplicity of $\lambda = g_\lambda = \dim V_\lambda = \dim(\ker(A - \lambda I)) = \text{null}(A - \lambda I)$.
3. Minimal multiplicity of $\lambda = m_\lambda =$ power of $(x - \lambda)$ in $g(x)$.

Definition 27. $A \in \mathbb{C}^{n \times n}$ is Diagonalizable iff there is an invertible matrix P and a diagonal matrix D such that $A = P^{-1}DP$. i.e. when A is similar to a diagonal matrix D .

Example 20. Find algebraic multiplicities, geometric multiplicities, and minimal polynomial of the following matrices and diagonalize A when possible.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 3 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

Theorem 31. The following are equivalent

1. $A \in \mathbb{C}^{n \times n}$ is diagonalizable
2. $a_\lambda = g_\lambda$
3. $m_\lambda = 1$
4. Eigenvectors form a basis for $\mathbb{C}^n$

Definition 28. $A \in \mathbb{C}^{n \times n}$ is Hermitian iff $A^H = A$ where $A^H = \overline{A}^T$

Theorem 32. Let $A \in \mathbb{C}^{n \times n}$ is Hermitian, then

1. eigenvalues are real
2. eigenvectors corresponds to distinct eigenvalues are orthogonal
3. (spectral theorem): it has n orthogonal($\Rightarrow$ orthonormal)eigenvectors.

Definition 29. Let $P \in \mathbb{C}^{n \times n}$ is said to be Unitary iff $P^H P = P P^H = I$ i.e. when $P^{-1} = P^H$.

If P has ortonormal columns then P is Unitary.

Definition 30. $A \in \mathbb{C}^{n \times n}$ is Positive Definite(PD) iff $x^H A x > 0$ for all $x \neq \underline{0}$

Theorem 33. If A is Hermitian then A is PD iff all its eigenvalues are positive.

Example 21. Use matrix diagonalization to solve the following problems

1. Identify the surface give by $f(x, y, z) = 2x^2 + 12xy + y^2 - 4xz - 8yz - 3z^2 = 0$

Hint: We can find A Hermitian such that

$$f = X^H A X = X^H P \Lambda P^H X = (P^H X)^H \Lambda (P^H X) = Y^H \Lambda Y \text{ where } Y = P^H X.$$

2. Solve the system first order ODEs $x' = 3x + y + t, y' = 2x + 4y + t^2$

Hint: $X' = AX + b(t) = P \Lambda P^{-1} X + b(t) \Rightarrow P^{-1} X' = \Lambda P^{-1} X + P^{-1} b(t)$

$\Rightarrow Y' = \Lambda Y + c(t)$, where $Y = P^{-1} X, c(t) = P^{-1} b(t)$

Definition 31. Jordan decomposition for $A \in \mathbb{C}^{n \times n}$

$A = P J P^{-1}$ where P is invertible and J consists of Jordan blocks on the diagonal and the rest of the elements are 0(if they exists).

Jordan block for λ consists of λ on the diagonal, 1 on the upper sub diagonal(if it exists) and rest of the elements are 0. Example

Jordan block if size 1: (λ)

Jordan block if size 2: $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$

Jordan block if size 3: $\begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$

Theorem 34.

1. Every $A \in \mathbb{C}^{n \times n}$ has a Jordan decomposition.
2. If $A \in \mathbb{C}^{n \times n}$ has n L.I. eigenvectors then the Jordan decomposition coincides with the Eigenvalue decomposition.

Example 22. Find the Jordan decomposition of the following matrices

$$\begin{pmatrix} 3 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 & 0 & 0 & 3 & 3 \\ 0 & 2 & 0 & 3 & 3 & 3 \\ 0 & 0 & 2 & 3 & 3 & 3 \\ 0 & 0 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

Theorem 35. If Jordan blocks of the eigenvalue are of the sizes λ are $1 \leq k_1, k_2, \dots, k_r$:

1. $g_\lambda = r$
2. $a_\lambda = \sum_{i=1}^r k_i \geq \sum_{i=1}^r 1 = r = g_\lambda$
3. $m_\lambda = \max\{k_1, k_2, \dots, k_r\} = k_j$, say
4. Therefore $a_\lambda - g_\lambda = \sum_{i=1}^r (k_i - 1) \geq k_j - 1 = m_\lambda - 1$, so $a_\lambda \geq g_\lambda$ since $m_\lambda \geq 1$
5. $a_\lambda - g_\lambda = m_\lambda - 1$ if the size of the matrix $n \leq 3$.

Definition 32. QR decomposition of $A \in \mathbb{C}^{m \times n}$ is

$A = QR$ where $Q \in \mathbb{C}^{m \times n}$ consists of orthonormalized columns of A by the Gram-Schmidt process and $R \in \mathbb{C}^{n \times n}$ is an uppertriangular matrix consists coefficients of the Gram-Schmidt process.

If A is not square or not having L.I columns, we can add more columns to Q and make it Square and Unitary. Then by adding rows or columns to R as necessary and still make $A = QR$ hold.

Theorem 36. QR algorithm for finding eigenvalues

Let A be square and $A = A_0 = Q_0 R_0$ be the QR decomposition of A . Define $R_{k-1} Q_{k-1} = A_k = Q_k R_k$ where $A_k = Q_k R_k$ be the QR decomposition of A_k for $k \geq 1$.

We see that $A_k = Q_k R_k Q_k^H = Q_k A_{k+1} Q_k^H$ and A_k and A_{k+1} have the same eigenvalues. Under suitable conditions, A_k approaches an upper triangular matrix, whose diagonal consists of eigenvalues of A .

Note 8. Following facts about the SVD decomposition was not covered in lectures or quizzes, however it consists of examples that you can try.

Theorem 37. If $A \in \mathbb{C}^{m \times n}$ then

1. $A^H A$ and AA^H are Hermitian and the eigenvalues are non negative.
2. $A^H A$ and AA^H have the same positive eigenvalues.
3. $\text{rank}(A^H A) = \text{rank}(AA^H) = \text{rank} A^H = \text{rank} A = \text{no. of positive eigenvalues of } A^H A \text{ or } AA^H$

Definition 33. Let $A \in \mathbb{C}^{m \times n}$

Singular values $\sigma \geq 0$ and the Singular vectors $u \in \mathbb{C}^{m \times 1}, v \in \mathbb{C}^{n \times 1}$ are defined by $Av = \sigma u$ and $A^H u = \sigma v$.

We obtain σ, u, v by the eigenvalue problem: $AA^H u = \sigma^2 u$ and $A^H A v = \sigma^2 v$.

There will be $\min\{m, n\}$ number of common eigenvalues $\sigma^2 \geq 0$ (which are called singular values) and $|m - n|$ number of $\sigma^2 = 0$ other eigenvalues.

If $\sigma^2 > 0$, we can select $\sigma > 0$.

Then we can write: $AV = U\Sigma$ where, matrices $V \in \mathbb{C}^{n \times n}$ consists of v 's and V is Unitary and $U \in \mathbb{C}^{m \times m}$ consists of u 's and U is Unitary and $\Sigma \in \mathbb{R}^{n \times m}$ consists of common eigen values on its diagonal and 0's elsewhere.

Finally we have the SVD(Singular Value Decomposition): $A = U\Sigma V^H$.

Theorem 38.

1. Every matrix $A \in \mathbb{C}^{m \times n}$ has a SVD.
2. $\text{rank}(A^H A) = \text{rank}(AA^H) = \text{rank} A = \text{rank} A^H = \text{no. of non zero singular values of } A$.